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Cascade, inheritance and specificity 45m

You may have already encountered a situation where you expected a specific CSS style to be applied to an element, but you instead got a different style. Often, what has happened is that you will have created two different styles to be applied to the same element – which style will be applied to the element in this case?

This is where the concepts of cascade, specificity and inheritance come in. They will control which style will be applied when more than one style applies. So, what are these concepts?

Watch *CSS: The cascade, specificity and inheritance* (VirtualAddiction, 2021) (15:00)^{fnref:4cc05c} to learn about the concepts of cascade, inheritance and specificity and how these concepts are used to define the order of importance of CSS styles. Though this video is quite long, it is one of the best out there.



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Cascade

Cascading is the process of combining rules. Where there is no conflict, all of the properties apply to the same selector. Should they conflict, the following rules are applied in order of weight: importance, specificity and source order.

Inheritance

Earlier, we learnt that HTML elements can be nested inside each other. For example, the `<body>` element encloses other elements (such as `<h1>` and `<p>` elements). The `<body>` element is considered the parent. The styling of parent elements is carried down to the child elements. Child elements inherit the style of the parent unless specifically overridden by their own styles.

Specificity

The less specific the elements, the more can be matched by the selector; the more specific, the fewer elements can be matched by the selector. The simplified rule for working out specificity is that the following order is applied from most specific to least specific: id selector, class selector or pseudo-class selector, and element type selector or pseudo-element selector.

Now that you understand what cascade, inheritance and specificity mean regarding CSS, how are these concepts used to determine the order in which the CSS rules are applied?

Watch *CSS beginner concepts – cascade & specificity* (Thirus, 2021)

(6:41),^{fnref:195b72} which demonstrates the order of importance through examples.



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Have a go at Task 1 and Task 2 of *Test Your Skills: The Cascade* (MDN, 2025).^{fnref:c86915}

See if you can change the colours of the font and background. Do not use a colour value to change the colours; see if you can change the colours using only inheritance and cascade rules.



☒ Understanding CSS inheritance and overriding rules

Non-Assessed

Status: Done

Reset ↺

Given the following HTML and CSS code, update the CSS such that the background colour of the `<a>` elements is inherited from the parent containers instead of being specifically defined.

Task: Understanding CSS inheritance and overriding rules

The `<a>` elements have a blue background explicitly set by the CSS rule `#outer div ul .nav a`. However, we want the `<a>` elements to inherit the background colour from their parent containers.

You are required to:

- Modify the CSS such that the `<a>` elements inherit the background colour of their parent containers. If no background colour is defined in the parent container, the `<a>` elements should have no background colour applied.

☒ I have successfully completed this task!



To achieve inheritance, you could write a CSS rule like this:

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```
#outer #inner a {  
  background-color: inherit;  
}
```

Adjust the other CSS rules as necessary to achieve the desired inheritance effect while maintaining styling consistency.

Final output:

1. The `<a>` elements inherit the background colour of their parent containers.
2. Parent containers should control the appearance of child elements via CSS inheritance, ensuring reusability of styles.

Previous attempts ▼



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Dig deeper

To learn more about cascade, inheritance and specificity:

- Read about cascade and inheritance, on the *Handling Conflicts* page which goes into detail about each concept and provides examples.
- Read or listen to *The Cascade* podcast to learn how a browser prioritises the cascade of competing CSS rules.

Let's now look at CSS fonts.



Thoughts on this page?

